



**FINAL EXAMINATION
SEMESTER II SESSION 2022/2023**

FACULTY OF INFORMATICS AND COMPUTING

DEGREE PROGRAMME

COURSE : DATABASE
COURSE CODE : CSF21303
DURATION : 2 HOURS AND 30 MINUTES

INSTRUCTIONS TO CANDIDATES

1. Answer ALL questions.
2. Answer all questions in the Answer Booklet provided. Begin each answer on a new page.

This Question Booklet consists of 6 printed pages.

Question 1 [20 Marks]

a) Derive many-to-many relationship in Figure 1 into composite entity.

[4 Marks]



Figure 1. Many-to-many relationship

b) Create the appropriate Chen Model for each of the specified relationship.

[8 Marks]

- i. A company operates four departments.
- ii. Each department employs employees.
- iii. Each of the employees may or may not have one or more dependents.
- iv. Each employee may or may not have an employment history.

c) Define the following terms.

[8 Marks]

- i. Domain
- ii. Tuple
- iii. Candidate key
- iv. Referential integrity

Question 2 [25 Marks]

a) The relationship structure is portrayed in two simple and common formats. Briefly describe these two formats.

[4 Marks]

b) Convert the Entity Relationship Diagram (ERD) in Figure 2 into a relational database schema. Indicate primary keys and referential integrity constraints in your schema.

[9 Marks]

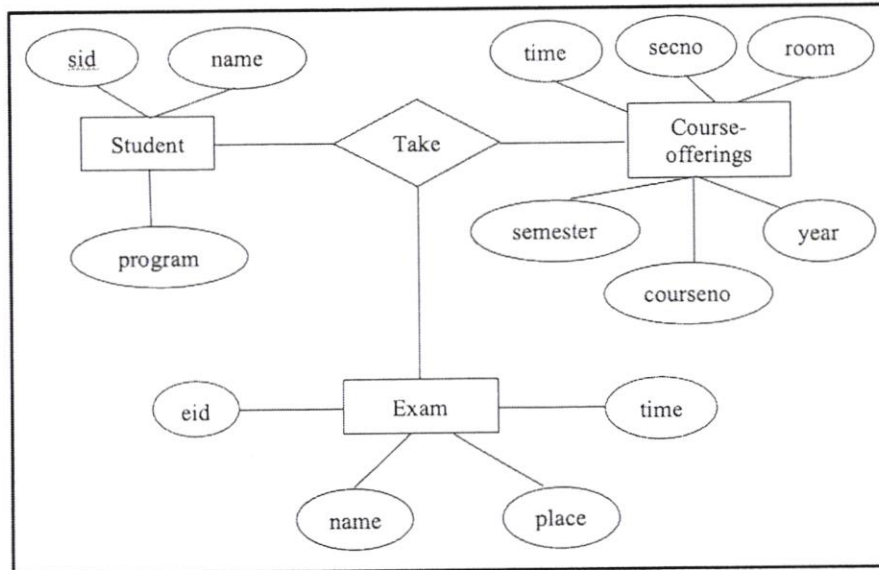


Figure 2. Entity Relationship Diagram

c) Transform superclass/subclass relationship in Figure 3 into table.

[8 Marks]

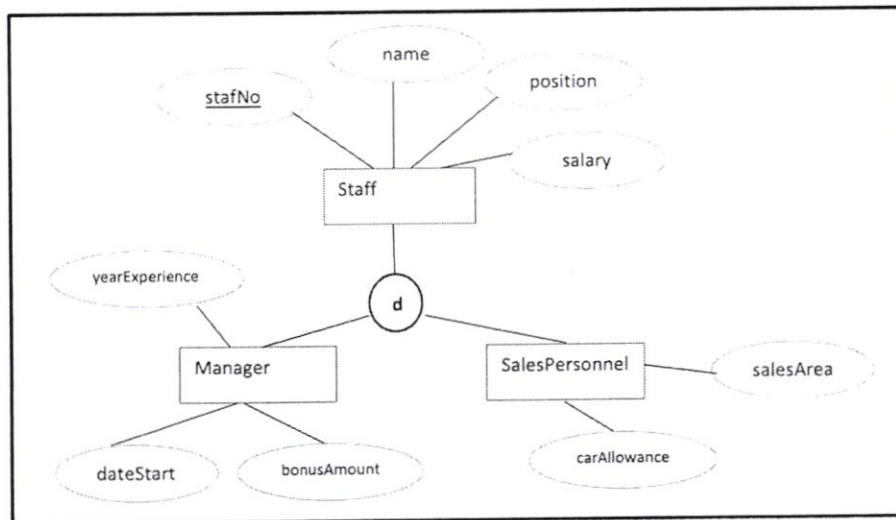


Figure 3. Superclass/subclass relationship

d) Naming the 'd' rule in Figure 3 and explain it.

[4 Marks]

Question 3 [30 Marks]

An art museum wants to keep track of its collection. The following requirements describe the information it wants to keep:

- The museum has a collection of ART_OBJECTs. Each ART_OBJECT has a unique IdNo, ArtistName, a YearCreated, a Title, and a Description. The art objects are categorized in several ways, as discussed below.
- ART_OBJECTs are categorized based on their type. There are three main types: PAINTING, SCULPTURE, and STATUE, plus another type called OTHER to accommodate objects that do not fall into one of the three main types.
- A PAINTING has a PaintType, material on which it is DrawnOn (paper, canvas, wood, etc.), and Style (modern, abstract, etc.).
- A SCULPTURE or a STATUE has a Material from which it was created (wood, stone, etc.), Height, Weight, and Style.
- An art object in the OTHER category has a Type (print, photo, etc.) and Style.
- ART_OBJECTs are also categorized as PERMANENT_COLLECTION, which are owned by the museum (these have information on the DateAcquired, whether it is OnDisplay or stored, and Cost) or BORROWED, which has information on the Collection (from which it was borrowed), DateBorrowed, and DateReturned.
- ART_OBJECTs also have information describing their country/culture using information on country/culture of Origin (Italian, Egyptian, American, Indian, etc.) and Epoch (Renaissance, Modern, Ancient, etc.).
- The museum keeps track of ARTIST's information, if known: Name, DateBorn (if known), DateDied (if not living), CountryOfOrigin, Epoch, MainStyle, and Description. The Name is assumed to be unique.

a) Produce an Entity Relationship Diagram (ERD) to represent the museum's database design.

[20 Marks]

b) Provide a data dictionary for each entity in (a).

[8 Marks]

c) Discuss the concept of specialization based on above requirements.

[2 Marks]

Question 4 [25 Marks]

a) Normalization is a process within a logical design. What is the general goal of normalization?

[2 Marks]

b) Based on the following Staff_Branch relation, explain **THREE (3)** types of anomalies in database design by using an appropriate example.

[12 Marks]

Table 1: Staff_Branch

StaffNo	SName	Position	Salary	BranchNo	BAddress
SL21	Adam	Manager	30000	B005	22 Jalan Budiman
SG37	Umar	Assistant	12000	B003	163 Taman Uda
SG14	Afiq	Supervisor	18000	B003	163 Taman Uda
SA9	Aisyah	Assistant	9000	B007	16 Jalan Pegawai
SG5	Ulfah	Manager	24000	B003	163 Taman Uda
SL41	Fatimah	Assistant	9000	B005	22 alan Budiman

c) Based on the CustSupplyProduct in Table 2, answer the question below.

Table 2: CustSupplyProduct

CustId	Name	Address	Product Id	ProductName	Quatity	Price	SupplierCode	SupplierAddress
PG001	Izara	Bangi	KB100	Papan Kekunci	20	25.00	Permata Sdn Bhd	Nilai
			CP230	CPU	10	700.00	Cemerlang Ent	Gombak
			MS112	Tetikus	3	15.00	Permata Sdn Bhd	Nilai
PG002	Warda	Shah Alam	HD200	Hard disk	10	140.00	Permata Sdn Bhd	Nilai
			KB100	Papan Kekunci	8	25.00	Permata Sdn Bhd	Nilai
			SP036	Pembesar suara	6	50.00	Cemerlang Ent	Gombak
PG003	Ejaz	Ampan g	MS112	Tetikus	4	15.00	Permata Sdn Bhd	Nilai
			CP230	CPU	3	700.00	Cemerlang Ent	Gombak
			VG400	Kad VGA	5	215.00	Cemerlang Ent	Gombak

- Describe why the table shown above is not in the First Normal Form (1NF). **[2 Marks]**
- Transform the relational table above into First Normal Form (1NF). **[3 Marks]**
- Transform the relational table above into Second Normal Form (2NF). **[3 Marks]**
- Transform the CustSupplyProduct relation into Third Normal Form (3NF). **[3 Marks]**

END OF QUESTION PAPER